

*****This Document involves the topics for Google Play Store Data Analysis Topics*****

Dataset- <https://www.kaggle.com/datasets/lava18/google-play-store-apps?select=googleplaystore.csv>

Basic-Level Topics:

1. What is the average rating of apps in the dataset?

- **Hint:** Use pandas to calculate the mean of the Rating column.
- **How it helps:** Understanding the general user satisfaction level with apps.
- **Business Impact:** Identifying the average satisfaction can help set a benchmark for new apps and improve quality standards.

2. How many unique categories of apps are there?

- **Hint:** Use pandas to find the unique values in the Category column.
- **How it helps:** Identifying the diversity of app categories available.
- **Business Impact:** Understanding the variety of app categories can help in identifying market opportunities and potential areas for new app development.

3. What is the distribution of app sizes?

- **Hint:** Use matplotlib or seaborn to create a histogram of the Size column.
- **How it helps:** Understanding the typical size of apps can help in resource allocation and development.
- **Business Impact:** Knowing the distribution of app sizes can guide infrastructure planning and optimization of app performance.

4. How many free vs paid apps are there?

- **Hint:** Use pandas to count the occurrences of each value in the Type column.
- **How it helps:** Gauging the market split between free and paid apps.
- **Business Impact:** Understanding the market dynamics between free and paid apps can inform pricing strategies and marketing campaigns.

5. What is the most common content rating for apps?

- **Hint:** Use pandas to find the mode of the Content Rating column.
- **How it helps:** Understanding the target audience for most apps.
- **Business Impact:** Knowing the prevalent content rating can help in designing apps that appeal to the largest user base and ensuring compliance with age-appropriate content guidelines.

6. What are the top 5 most installed apps?

- **Hint:** Sort the dataset by the Installs column and select the top 5.
- **How it helps:** Identifying popular apps can inform marketing and development strategies.
- **Business Impact:** Highlighting the most installed apps can showcase successful case studies and strategies that can be emulated to achieve similar success.

7. How many apps have a rating of 4.0 and above?

- **Hint:** Use pandas to filter the Rating column for values ≥ 4.0 .
- **How it helps:** Focusing on high-quality apps can highlight successful development practices.
- **Business Impact:** Identifying high-rated apps can help in understanding best practices and setting benchmarks for new app developments to achieve high user satisfaction.

8. What is the average number of reviews for free vs paid apps?

- **Hint:** Group the data by Type and calculate the mean of the Reviews column.
- **How it helps:** Assessing user engagement with free versus paid apps.
- **Business Impact:** Understanding the engagement levels can inform promotional strategies and product positioning to maximize user feedback and engagement.

9. What is the average app size for each category?

- **Hint:** Group the data by Category and calculate the mean of the Size column.
- **How it helps:** Understanding category-specific size requirements can guide development.
- **Business Impact:** Knowing the average size for each category can help optimize resource allocation and improve app performance specific to each category.

10. How many apps were last updated in 2018?

- **Hint:** Convert Last Updated to datetime and filter for the year 2018.
- **How it helps:** Analyzing update frequency can inform maintenance strategies.
- **Business Impact:** Understanding update patterns can help in planning regular updates and maintaining app relevance and user engagement over time.

Medium-Level Topics:

1. What is the correlation between the number of installs and the app rating?

- **Hint:** Use pandas and numpy to calculate the correlation between the Installs and Rating columns.
- **How it helps:** Understanding the relationship between popularity and user satisfaction.
- **Business Impact:** Identifying if higher installs lead to better ratings can guide marketing and user acquisition strategies.

2. Which app categories have the highest average rating?

- **Hint:** Group the data by Category and calculate the mean of the Rating column.
- **How it helps:** Identifying top-performing categories based on user ratings.
- **Business Impact:** Focusing on high-rated categories can guide investment and development decisions to enhance user satisfaction.

3. How does the price of an app affect its average rating?

- **Hint:** Group the data by Price and calculate the mean of the Rating column. Filter for paid apps only.
- **How it helps:** Understanding if more expensive apps tend to have higher or lower ratings.
- **Business Impact:** Informing pricing strategies to optimize user satisfaction and revenue.

4. What is the distribution of app ratings across different content ratings?

- **Hint:** Use seaborn to create a boxplot of Rating against Content Rating.

- **How it helps:** Visualizing how ratings vary with content appropriateness.
- **Business Impact:** Tailoring app content and marketing strategies based on content rating performance.

5. Which genres have the most apps with over 1 million installs?

- **Hint:** Filter the data for apps with Installs greater than 1 million and then group by Genres.
- **How it helps:** Identifying popular genres among high-install apps.
- **Business Impact:** Guiding genre-specific development and marketing efforts to capitalize on high-install trends.

6. How frequently do apps get updated? Calculate the average time between updates.

- **Hint:** Convert Last Updated to datetime and calculate the difference between consecutive updates for each app.
- **How it helps:** Understanding the update cycle of apps.
- **Business Impact:** Planning regular updates to maintain app relevance and user engagement.

7. What is the impact of app size on the number of installs?

- **Hint:** Create a scatter plot using matplotlib or seaborn with Size and Installs.
- **How it helps:** Analyzing if app size affects download popularity.
- **Business Impact:** Optimizing app size to balance performance and user appeal.

8. Which apps have the highest number of reviews, and what are their ratings?

- **Hint:** Sort the dataset by Reviews and select the top apps, then check their Rating.
- **How it helps:** Identifying highly reviewed and rated apps can provide insights into user preferences.
- **Business Impact:** Learning from highly reviewed apps to improve app features and user engagement.

9. How does the content rating distribution differ between free and paid apps?

- **Hint:** Use pandas to create a crosstab of Content Rating and Type.
- **How it helps:** Understanding the target audience for free vs paid apps.
- **Business Impact:** Developing targeted marketing strategies based on content rating and app type.

10. What are the top 5 categories with the most installs?

- **Hint:** Group the data by Category and sum the Installs column, then sort and select the top 5.
- **How it helps:** Identifying categories with high user demand.
- **Business Impact:** Prioritizing development and marketing efforts in categories with the highest install rates.

Advanced-Level Topics:

1. What are the top 10 apps with the highest ratings, and how do their number of reviews and installs compare?

- **Hint:** Filter the dataset for apps with the highest ratings, then compare their Reviews and Installs.
- **How it helps:** Understanding if the highest-rated apps also have high user engagement and popularity.
- **Business Impact:** Identifying successful apps to benchmark new app development and marketing strategies.

2. Analyze the trend of app updates over time. Are there any noticeable patterns or seasonal trends?

- **Hint:** Convert Last Updated to datetime and plot the frequency of updates over time using matplotlib or seaborn.
- **How it helps:** Identifying patterns in app updates can reveal industry practices or seasonal effects.
- **Business Impact:** Planning update schedules to align with industry trends and maximize user engagement.

3. How does the average rating of apps change with the number of installs? Create a binned analysis.

- **Hint:** Create bins for Installs and calculate the average Rating for each bin.
- **How it helps:** Understanding how user ratings vary with app popularity.
- **Business Impact:** Tailoring user acquisition strategies based on how ratings evolve with the user base.

4. Perform sentiment analysis on app reviews (if review text is available) to determine the common themes in high and low-rated apps.

- **Hint:** Use NLP techniques in Python to analyze the sentiment of review texts and categorize themes.
- **How it helps:** Gaining deeper insights into user feedback and common issues or praises.
- **Business Impact:** Addressing common complaints and enhancing features that users love to improve app ratings.

5. What is the relationship between app genre and user ratings? Are certain genres consistently rated higher or lower?

- **Hint:** Group the data by Genres and calculate the mean and median of the Rating column.
- **How it helps:** Identifying genre-specific trends in user satisfaction.
- **Business Impact:** Focusing on high-rated genres for new app development to align with user preferences.